Inverse Functions and Implicit Differentiation

MATH 1334 October 21, 2026

3.7 Derivatives of inverse functions

Question. How do we find $\dfrac{d}{dx} f^{-1}(x)$?

An inverse function *undoes* the original function:

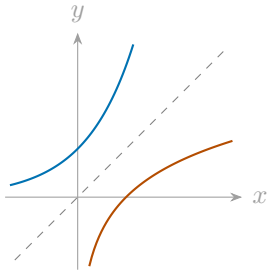
$$f^{-1}(f(x)) = x, \qquad f(f^{-1}(y)) = y.$$

Here f must be one-to-one on the domain being used. The notation f^{-1} means the inverse function, not the reciprocal $1/f$.

The graphs of f and f^{-1} are reflections across $y = x$. The domain and range exchange roles. The dashed line in each sketch is $y = x$.

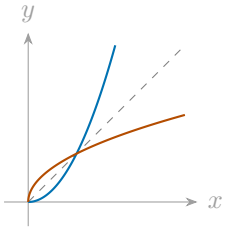
Exponential and logarithm

$$f(x) = e^x, \quad f^{-1}(x) = \ln x.$$



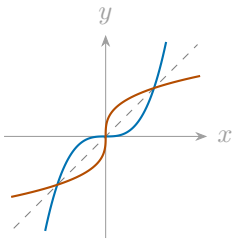
Square and square root

$$f(x) = x^2 \ (x \geq 0), \quad f^{-1}(x) = \sqrt{x}.$$



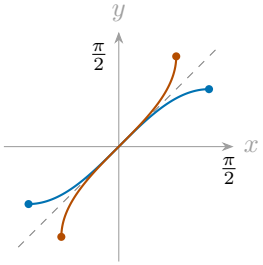
Cube and cube root

$$f(x) = x^3, \quad f^{-1}(x) = \sqrt[3]{x}.$$



Sine and inverse sine

$$f(x) = \sin x \quad (-\tfrac{\pi}{2} \leq x \leq \tfrac{\pi}{2}),$$
$$f^{-1}(x) = \sin^{-1} x = \arcsin x.$$



The inverse-function derivative

Theorem. *Suppose f is one-to-one and continuous on an open interval containing a, is differentiable at a, and $f'(a) \neq 0$. If $b = f(a)$, then f^{-1} is differentiable at b, and*

$$(f^{-1})'(b) = \frac{1}{f'(a)} = \frac{1}{f'(f^{-1}(b))}.$$

Proof. Let $b = f(a)$, so $a = f^{-1}(b)$. Apply the chain rule to

$$f^{-1}(f(x)) = x :$$

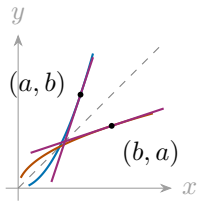
$$(f^{-1})'(f(x))\, f'(x) = 1.$$

At $x = a$, this gives $(f^{-1})'(b)\, f'(a) = 1$. Since $f'(a) \neq 0$,

$$(f^{-1})'(b) = \frac{1}{f'(a)} = \frac{1}{f'(f^{-1}(b))}.$$ □

Geometrically. Reflection across $y = x$ interchanges rise and run. Tangent slopes at (a, b) and (b, a) are reciprocals:

$$m_{f^{-1}} = \frac{1}{m_f}.$$



Example 1. Suppose $f(2) = 5$, $f'(2) = -3$, and f satisfies the theorem's hypotheses. Compute $(f^{-1})'(5)$.

$$f^{-1}(5) = 2, \qquad (f^{-1})'(5) = \frac{1}{f'(f^{-1}(5))} = \frac{1}{f'(2)} = -\frac{1}{3}.$$

Example 2. Let $f(x) = x^2$ for $x \geq 0$. Since $f(2) = 4$,

$$(f^{-1})'(4) = \frac{1}{f'(2)} = \frac{1}{2 \cdot 2} = \frac{1}{4}.$$

Example 3. Let $f(x) = x^3$. Since $f(2) = 8$,

$$(f^{-1})'(8) = \frac{1}{f'(2)} = \frac{1}{3 \cdot 2^2} = \frac{1}{12}.$$

Check. $f^{-1}(x) = x^{1/3}$, so $(f^{-1})'(8) = \frac{1}{3}\, 8^{-2/3} = \frac{1}{3} \cdot \frac{1}{4} = \frac{1}{12}$.

Derivatives of inverse trigonometric functions

Proposition. *With the usual inverse-trigonometric branches, the following derivatives hold on the indicated open domains. Angles are measured in radians.*

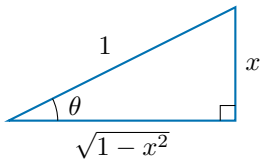
$$\begin{aligned}
&(i) && \frac{d}{dx}\sin^{-1}x &&= \frac{1}{\sqrt{1-x^2}}, && |x|<1,\\
&(ii) && \frac{d}{dx}\cos^{-1}x &&= -\frac{1}{\sqrt{1-x^2}}, && |x|<1,\\
&(iii) && \frac{d}{dx}\tan^{-1}x &&= \frac{1}{1+x^2}, && x\in\mathbb{R},\\
&(iv) && \frac{d}{dx}\csc^{-1}x &&= -\frac{1}{|x|\sqrt{x^2-1}}, && |x|>1,\\
&(v) && \frac{d}{dx}\sec^{-1}x &&= \frac{1}{|x|\sqrt{x^2-1}}, && |x|>1,\\
&(vi) && \frac{d}{dx}\cot^{-1}x &&= -\frac{1}{1+x^2}, && x\in\mathbb{R}.
\end{aligned}$$

Here $\cot^{-1}$ takes values in $(0,\pi)$, $\sec^{-1}$ in $[0,\pi]\setminus\{\pi/2\}$, and $\csc^{-1}$ in $[-\pi/2,\pi/2]\setminus\{0\}$.

Proof of (i). Let $\theta=\sin^{-1}x$. Then $\sin\theta=x$ and $-\pi/2<\theta<\pi/2$. By the inverse-function derivative theorem,

$$\frac{d}{dx}\sin^{-1}x=\frac{1}{\cos(\sin^{-1}x)}=\frac{1}{\cos\theta}.$$

To find $\cos\theta$, first draw a right triangle for $0<x<1$. Since $\sin\theta=x/1$, take the opposite side to be x and the hypotenuse to be 1.



The Pythagorean theorem gives the adjacent side $\sqrt{1-x^2}$, so

$$\cos\theta=\frac{\text{adjacent}}{\text{hypotenuse}}=\frac{\sqrt{1-x^2}}{1}=\sqrt{1-x^2}.$$

This also holds for all $-1<x<1$: $\cos^2\theta=1-\sin^2\theta=1-x^2$, and $\cos\theta>0$ for $-\pi/2<\theta<\pi/2$. Therefore

$$\frac{d}{dx}\sin^{-1}x=\frac{1}{\sqrt{1-x^2}}.$$

□

Using inverse-trigonometric derivatives

Example. Find $\dfrac{d}{dx}\cos^{-1}\sqrt{2x}$.

Chain rule. Differentiate the layers from outside inward: inverse cosine, square root, then $2x$.

$$\begin{aligned}\frac{d}{dx}\cos^{-1}\sqrt{2x} &= -\frac{1}{\sqrt{1-(\sqrt{2x})^2}}\,\frac{1}{2\sqrt{2x}}\,2\\ &= -\frac{1}{\sqrt{1-2x}\sqrt{2x}}\\ &= -\frac{1}{\sqrt{2x-4x^2}}, \qquad 0<x<\frac{1}{2}.\end{aligned}$$

The function is defined for $0 \le x \le \frac{1}{2}$; the displayed derivative applies in the interior.

Practice

1. Find $\dfrac{d}{dx}(\tan^{-1}\sqrt{x})$.

$$\begin{aligned}\frac{d}{dx}\tan^{-1}\sqrt{x} &= \frac{1}{1+(\sqrt{x})^2}\,\frac{1}{2\sqrt{x}}\\ &= \frac{1}{2\sqrt{x}(1+x)}, \qquad x>0.\end{aligned}$$

2. Find $y'(2)$, where $x\cos y = 2$.

For the principal inverse-cosine branch, write

$$y = \cos^{-1}\left(\frac{2}{x}\right).$$

Near $x = 2$, this branch has $x \ge 2$. For $x > 2$,

$$y' = \left(-\frac{1}{\sqrt{1-4/x^2}}\right)\left(-\frac{2}{x^2}\right) = \frac{2}{x\sqrt{x^2-4}}.$$

As $x \to 2^+$, $y' \to +\infty$. Thus $y'(2)$ does not exist as a finite derivative; the curve has a vertical tangent at $(2, 0)$.

The same conclusion from the equation. At $x = 2$, we have $\cos y = 1$, so $y = 2k\pi$. If a finite derivative existed, differentiating would give

$$\cos y - x\sin y\, y' = 0,$$

which would reduce to $1 = 0$. The vertical tangent is $x = 2$ at each point $(2, 2k\pi)$.

3.8 Implicit differentiation

An *explicit* formula writes y directly as a function of x:

$$y = f(x), \qquad y' = \frac{df}{dx}.$$

An *implicit* equation relates x and y:

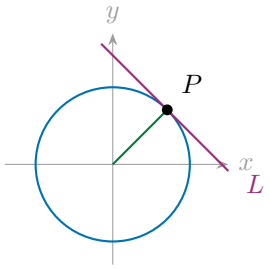
$$F(x, y) = 0.$$

For example, $x^2 + y^2 = 1$ gives the two branches $y = \pm\sqrt{1 - x^2}$, while $x \cos y = 2$ gives a different relation between x and y.

Question. How do we compute y' without first solving for y?

Example 1. Find the slope of the tangent line to $x^2 + y^2 = 1$ at $P = (1/\sqrt{2}, 1/\sqrt{2})$.

Geometry. The radius through P lies on $y = x$, so its slope is 1. The tangent is perpendicular to the radius, hence its slope is -1.



Method 1: solve for y. On the first-quadrant branch,

$$y = \sqrt{1 - x^2}, \qquad y' = -\frac{x}{\sqrt{1 - x^2}}.$$

At $x = 1/\sqrt{2}$,

$$y' = -\frac{1/\sqrt{2}}{\sqrt{1 - 1/2}} = -1.$$

Method 2: implicit differentiation. Treat y as a function of x and differentiate both sides:

$$\begin{aligned} \frac{d}{dx}(x^2 + y^2) &= \frac{d}{dx}(1), \\ 2x + 2y\,y' &= 0, \\ y' &= -\frac{x}{y} \qquad (y \neq 0). \end{aligned}$$

At P, $y' = -(1/\sqrt{2})/(1/\sqrt{2}) = -1$ again.

Implicit differentiation: Differentiate the equation with respect to x, using the chain rule whenever y appears, then solve for y'. The expression for y' may involve both x and y.

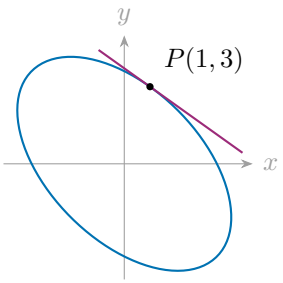
Tangent slopes and vertical tangents

Example 2. Find the slope of the tangent to $x^2 + xy + y^2 = 13$ at $P = (1, 3)$.

$$2x + \underbrace{y + xy'}_{\text{product rule}} + 2yy' = 0,$$

$$(x + 2y)y' = -2x - y,$$

$$y' = \frac{-2x - y}{x + 2y}.$$

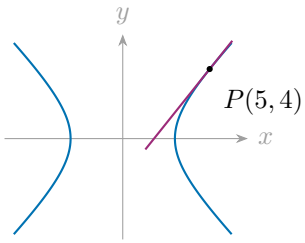


At $(1, 3)$, the slope is

$$y' = \frac{-2 - 3}{1 + 6} = -\frac{5}{7}.$$

Example 3. Find the slope of the tangent to $x^2 - y^2 = 9$ at $P = (5, 4)$.

$$2x - 2yy' = 0 \quad \Longrightarrow \quad y' = \frac{x}{y}.$$



At $(5, 4)$, the slope is $\dfrac{5}{4}$.

Example 4. For $x^2 - y^2 = 9$, find all points where the tangent is vertical.

A horizontal tangent has $dy/dx = 0$. For a vertical tangent, instead regard x as a function of y and look for $dx/dy = 0$.

$$\frac{d}{dy}(x^2 - y^2) = \frac{d}{dy}(9),$$

$$2x\frac{dx}{dy} - 2y = 0,$$

$$\frac{dx}{dy} = \frac{y}{x}.$$

The curve has $x \neq 0$, so $dx/dy = 0$ exactly when $y = 0$. Substituting in the original equation gives

$$x^2 = 9 \quad \Longrightarrow \quad x = \pm 3.$$

Thus the vertical tangents occur at $(3, 0)$ and $(-3, 0)$; their equations are $x = 3$ and $x = -3$.

Example: the curve $x^2 + xy + y^2 = 3$

1. Find the tangent slope at $P = (1, 1)$.

$$\begin{aligned} 2x + y + xy' + 2yy' &= 0, \\ (x + 2y)y' &= -2x - y, \\ y' &= \frac{-2x - y}{x + 2y}. \end{aligned}$$

At $(1, 1)$, $y' = (-2 - 1)/(1 + 2) = -1$.

2. Find all points where the tangent is horizontal.

Set $y' = 0$ in the differentiated equation. Then $2x + y = 0$, so $y = -2x$. Substitute into the curve:

$$x^2 + x(-2x) + (-2x)^2 = 3 \quad \Longrightarrow \quad 3x^2 = 3 \quad \Longrightarrow \quad x = \pm 1.$$

Find y in each case using $y = -2x$:

$$\begin{aligned} x = 1: \quad y = -2(1) = -2 \quad &\Longrightarrow \quad (1, -2), \\ x = -1: \quad y = -2(-1) = 2 \quad &\Longrightarrow \quad (-1, 2). \end{aligned}$$

The points are $(1, -2)$ and $(-1, 2)$. At both, $x + 2y = -3x \neq 0$, so the tangent slope is indeed zero.

3. Find all points where the tangent is vertical.

Differentiate with respect to y:

$$2x\frac{dx}{dy} + y\frac{dx}{dy} + x + 2y = 0, \qquad \frac{dx}{dy} = -\frac{x + 2y}{2x + y}.$$

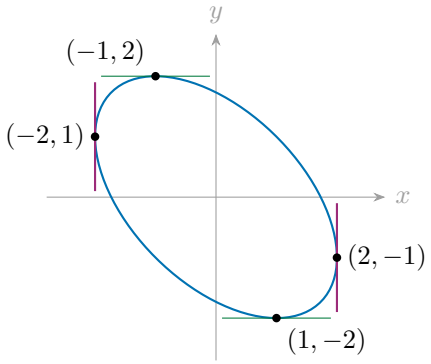
For $dx/dy = 0$, we need $x = -2y$. Substitution gives

$$(-2y)^2 + (-2y)y + y^2 = 3 \quad \Longrightarrow \quad 3y^2 = 3 \quad \Longrightarrow \quad y = \pm 1.$$

Find x in each case using $x = -2y$:

$$\begin{aligned} y = 1: \quad x = -2(1) = -2 \quad &\Longrightarrow \quad (-2, 1), \\ y = -1: \quad x = -2(-1) = 2 \quad &\Longrightarrow \quad (2, -1). \end{aligned}$$

Thus the vertical tangents occur at $(-2, 1)$ and $(2, -1)$. At both points, $2x + y = -3y \neq 0$.



Practice: the curve $x^2 - xy + y^2 = 12$

1. Find the tangent slope at $P = (2, -2)$.

$$\begin{aligned} 2x - y - xy' + 2yy' &= 0, \\ (-x + 2y)y' &= -2x + y, \\ y' &= \frac{-2x + y}{-x + 2y}. \end{aligned}$$

At $(2, -2)$,

$$y' = \frac{-4 - 2}{-2 - 4} = 1.$$

2. Find all points where the tangent is horizontal.

Set $y' = 0$. Then $2x - y = 0$, so $y = 2x$. Substituting,

$$x^2 - x(2x) + (2x)^2 = 12 \quad \Longrightarrow \quad 3x^2 = 12 \quad \Longrightarrow \quad x = \pm 2.$$

Find y in each case using $y = 2x$:

$$\begin{aligned} x = 2: &\quad y = 2(2) = 4 &&\Longrightarrow \quad (2, 4), \\ x = -2: &\quad y = 2(-2) = -4 &&\Longrightarrow \quad (-2, -4). \end{aligned}$$

Hence the horizontal tangents occur at $(2, 4)$ and $(-2, -4)$. The denominator $-x + 2y = 3x$ is nonzero at both points.

3. Find all points where the tangent is vertical.

Differentiate with respect to y:

$$2x\frac{dx}{dy} - y\frac{dx}{dy} - x + 2y = 0, \qquad \frac{dx}{dy} = \frac{x - 2y}{2x - y}.$$

For $dx/dy = 0$, we need $x = 2y$. Substituting,

$$(2y)^2 - (2y)y + y^2 = 12 \quad \Longrightarrow \quad 3y^2 = 12 \quad \Longrightarrow \quad y = \pm 2.$$

Find x in each case using $x = 2y$:

$$\begin{aligned} y = 2: &\quad x = 2(2) = 4 &&\Longrightarrow \quad (4, 2), \\ y = -2: &\quad x = 2(-2) = -4 &&\Longrightarrow \quad (-4, -2). \end{aligned}$$

Thus the vertical tangents occur at $(4, 2)$ and $(-4, -2)$. The denominator $2x - y = 3y$ is nonzero at both points.

